Rashi Bhave

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Education

- **Carnegie Mellon University** — Pittsburgh, PA
  M.S. Intelligent Information Systems (MIIS), Language Technologies Institute, SCS — *Aug 2026 – May 2028 (Expected)*
  Teaching Assistant: 95-820 Applications of NLP and LLMs; 94-804 AI Strategy (Heinz College)

- **Maharashtra Institute of Technology - WPU** — Pune, India
  B.Tech. Computer Science and Engineering — *Aug 2019 – May 2023*

Experience

- **Bravebird Inc.** — San Francisco, CA (Remote)
  Founding Machine Learning Engineer — *Feb 2026 – Jun 2026*
  - **Self-Healing RPA**: Designed and shipped the production **RPA** system for insurance **benefit verification**, automating **26 payer portals without triggering a single bot-detection block**, and cutting manual effort by an estimated **70%**; engineered **self-healing** logic recovering from UI and selector changes at runtime.
  - **Computer-Use & Voice Agents**: Developed an LLM-based **Computer-Use Agent (CUA)** pairing planner and executor agents to autonomously navigate payer portals and structure coverage details; architected a real-time **voice agent** for IVR-driven payer calls.

- **Nurix AI** — Bengaluru, India
  Machine Learning Engineer (Speech & GenAI) — *Sep 2024 – Jan 2026*
  - **In-House ASR (Voxtral)**: Curated **500 hours** of training data and fine-tuned **Voxtral** for Hindi–English code-switched speech, **improving WER by 22%** vs. Deepgram and AWS Transcribe.
  - **ML In Production**: Co-developed a real-time **voice agent** prototype for Nurix's first client, supporting **code-switching** over sensitive financial data; production architecture scaled to **800 calls/minute**.
  - **Speech Stack & RAG**: Engineered core **ASR**, **VAD**, and **TTS** components plus **PII redaction** for regulated deployments, and designed **Retrieval** algorithms later extended to multiple client deployments.

- **Microsoft Research India** — Bengaluru, India
  Research Fellow — *Jul 2023 – Aug 2024*
  - **Agentic Diagnostics (ACM FSE 2024)**: Designed and evaluated **ReAct**-style **LLM agents** with retrieval tools dynamically collecting telemetry for **Root Cause Analysis** in Azure data centers, cutting hallucination rate from **40% to 4%** on out-of-distribution production incidents.
  - **Confidence Estimation (RL)**: Implemented a **reinforcement-learning** pipeline around **Phi-3** to estimate confidence for suggested RCAs, **reducing calibration cost by 84%** across Azure teams.

Publications

- **Exploring LLM-Based Agents for Root Cause Analysis** — **ACM FSE 2024**
  *Devjeet Roy, Xuchao Zhang, **Rashi Bhave**, Chetan Bansal, et al.*

Selected Projects

- **MEDAI – Multimodal Healthcare AI**: Developed a health platform on **Meta Llama**, **DINOv2**, and **Detectron2** with AI-assisted pose correction, real-time fall detection (∼92% precision), and hybrid on-device + cloud inference (<200ms edge alerts).
- **Distributed Article Summarizer at Scale**: Architected a horizontally scalable **hierarchical summarization** system over millions of articles – **Qwen-2.5** on **vLLM**, **Weaviate** vector DB, **FastAPI**/**React** RAG chatbot deployed on **AWS** via **Terraform**.
- **SAARTHI (Hackathon Winner)**: Led a multi-agent system orchestrating **voice** (Whisper), **vision** (GPT-4o / LLaVA), and **chat** agents over a shared LangGraph planner for end-to-end business workflows; won **first place** at Nurix's agents-for-business hackathon.

Technical Skills

- **Languages**: Python (primary), C++, SQL, JavaScript / TypeScript, Bash
- **ML & GenAI**: LLM Agents, Multi-Agent Systems, Computer-Use Agents, RAG, Fine-Tuning, RL / RLHF, Distillation, Quantization, ASR / TTS / VAD, Code-Switched Speech, Computer Vision
- **Frameworks & MLOps**: PyTorch, Hugging Face, LangChain, LangGraph, vLLM, Triton, FastAPI, Docker, Terraform, AWS, MLflow, Git